STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 12: Kinetic Theory

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. What is an ideal gas model? [1]
2. Ideal gas equation? [1]
3. Mean translational kinetic energy? [1]
4. Origin of gas pressure? [1]
5. Monatomic degrees of freedom? [1]
6. Effect of lowering pressure on mean free path? [1]
7. Molar C_v of monatomic gas? [1]
8. Rms speed relation? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. At fixed T, volume halves: pressure? [2]
10. 27 deg C in Kelvin? [2]
11. Mean kinetic energy at 300K, k=1.38x10^-23? [2]
12. Temperature becomes 4T: rms speed factor? [2]
13. Internal energy of n mol monatomic gas? [2]
14. Two main ideal gas assumptions? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. Rms speed of M=.028,T=300K,R=8.31? [3]
16. 1 mol monatomic gas warms 20K, C_v=3R/2: DeltaU? [3]
17. At fixed T, pressure triples: volume factor? [3]
18. State equipartition prediction for monatomic gas. [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Compare rms speeds for molar masses .004 and .016 kg/mol at same T. [4]
20. f=5, n=2 mol, T=300K: internal energy? [4]

End of paper - check working, units, signs and significant figures.